Mathematical overview: pairs, copying and an ending

The all-paired starts are exactly those where the second player can always win. With finite counts in one, two or three ordered buckets, use the printed rule: remove one counter, optionally add at most one to each smaller-numbered bucket, and win by leaving all buckets empty. Empty buckets count as paired. From a start with every bucket even, the second player can copy the first player's move: take from the same bucket and refill exactly the same smaller buckets. The chosen bucket had at least two counters before the first take, so the copy is legal. Two takes and two of each refill leave every bucket paired again. As every game ends, the copier takes the last counter.

If any bucket is odd, the first player can leave every bucket even: take from the highest-numbered odd bucket and refill each smaller odd bucket once, leaving the smaller even buckets alone. Then copy. Under this one-each rule that first move is unique. The adult need not name parity or teach the rule before the search. "Can always win" means having a reply to *every* legal opponent move, not winning one trial.

Games end even when the buckets gain counters. For adults, write the counts as (a, b, c) in bucket order. Every move removes one counter, and every starting counter must go: the shortest game has $a + b + c$ moves, attained by never refilling. The nonnegative weight $a + 2b + 4c$ falls by at least one each turn, and by exactly one when every allowed refill is made. Thus the longest has $a + 2b + 4c$ moves. Omit c for two buckets. These bounds use the one-each cap; the upper table's uncapped game has different length claims.

What children discover. Problems 1–5: one bucket favors the second player exactly when it makes pairs; two buckets do so exactly when *both* make pairs. A child can say, "Whatever you do, I do the same thing to the same bucket. Everything is in twos again. There is always a move left for me, so you run out first." The first player wins by making pairs first. Problems 6–7 find every paired start, or show none exists. In cooperative Problems 8, 9 and 11, never refilling is shortest; each bucket-2 counter can earn one extra turn, each bucket-3 counter three. The board grows by at most one per bucket-3 take. Problem 10 restores pairs in three buckets; Problem 12 explains why every game ends.

Evidence. Playing is an experiment; the pairing rule is a conjecture until a child explains a reply to *any* move. The last bucket cannot refill; after its last take, the earlier buckets give a smaller game. The original guide's overview (p. 1) gives the older bands' destination, including finite but unbounded uncapped play; keep that rule separate.

Preparation and access

Status. Unpiloted with children. Printed counter fit, cutting, masking, drawing space, handling, adult workload and suggested timings are physically unrehearsed.

Print and kit. One seven-page `k-1.pdf` packet per child; each pair shares the bucket board (W81-S-v3, p. 2 of `week-81-students.pdf`): three 56×139 mm buckets with 5 mm gaps, and 86×40 mm Supply and Removed boxes. Print US Letter, single-sided, Actual Size. Use about 30 loose counters plus a reserve; choose flat counters no wider than about 20 mm and test that 11 counters fit flat in one bucket (Problem 1). Add a pencil and scrap paper to cover unused buckets. Small worksheet pictures are setup instructions, not counter mats. The week's sixteen 39×27.5 mm cut cards (shared packet p. 3) can stay in the kit but are not needed for this first route. No new printable object is missing.

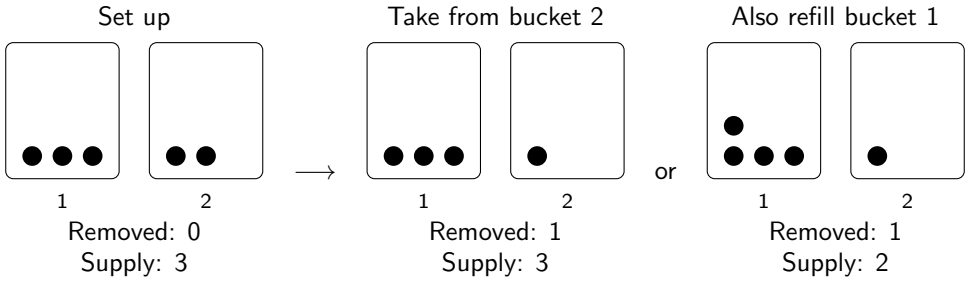
Beyond the week's kit: allow about 5–10 minutes to print/stack the K–1 pages, prepare masks and check a reset. This is an estimate, not a rehearsal. Use two pair kits for the table.

Launch, pacing and the first games

Prerequisites and staffing. One anchored adult for the K–1 table reads each prompt and the 1st/2nd choices aloud, records a child's chosen result when asked and checks resets. Children count small sets, recognize bucket labels 1–3, take turns and match dots to counters. The core reaches 11 counters; optional length work reaches 12 moves, while Problem 12 needs no count to 14. Children need not know "even; pairing counters is how they see it. Children write numerals up to 12 in Problems 8, 9 and 11, or dictate for the adult to write. Problems 5–6 ask for small bucket drawings; allow dictated dots or enlarged scrap sketches. The child still chooses the bucket, whether to refill and what to try next. If adults must choose moves, stay with one bucket and a few contrasting games. With an odd child, rotate player, player and checker; the adult can be a temporary partner.

Join the common launch. After the movement opening and material handling, use the week's short demonstration: remove one from the middle bucket, then optionally add one to bucket 1. The partner points to the chosen bucket before removal and checks additions. Refill only smaller-numbered buckets, from Supply.

At the table, about three minutes. Match the first rule picture on student p. 1 to the large board: three counters in bucket 1 and two in bucket 2, with bucket 3 covered and Removed empty. Take one from bucket 2 into Removed; any refill comes from Supply. Show both possible endings, resetting before the second. Neither is a winning first move for this non-task start, so no task answer is supplied.



Draw dots only for the *resulting buckets*, not arrows for hand movements. Point to the matching before/after pictures on p. 1; children may dictate or show a result rather than draw it. Cover buckets 2 and 3 and put just one counter in bucket 1, a non-task start. The first player takes it; the second cannot move, so the first wins. Point to the "1st / 2nd choices in Problem 1 and say which label would be circled for this example. Reset buckets and empty Removed before every game.

A first hour. Allow 5 minutes movement, 5 minutes handling/common launch, 3 minutes table setup, 10 minutes Problem 1, 15 minutes Problems 2–3, 15 minutes selected Problem 4 games or Problem 5 (rotation below), and 7 minutes sharing a move and a question. Choose only a few starts at first; do not require every game in both orders. Save Problems 6–7 for children ready to organize all splits, and Problems 8–12 for a later visit or a fluent pair. Finishing every page is not the goal. Record rule errors and who chose the moves, not only wins.

Key notation. (a, b) lists the counters in buckets 1 and 2; children can use dots.

Problem 1 (student p. 1): one bucket

In printed left-to-right order:

2	5	3	6	4	7	11	10
2nd	1st	1st	2nd	2nd	1st	1st	2nd

There is exactly one move each turn, so the game lasts as many turns as the start has counters. Two counters give one turn to each player; an unpaired counter gives the first player the last turn. *If needed:* "Can these counters all have partners? Whose turn takes the leftover?"

Winning moves and complete collections

Problem 2 (student p. 2): who can always win?

Printed across the top row, then across the bottom:

$(1,1)$	$(2,0)$	$(0,2)$	$(2,1)$	$(1,2)$	$(2,2)$
1st	2nd	2nd	1st	1st	2nd

A single win is not a guarantee. Let a partner challenge a proposed strategy with a different legal move. For $(2,2)$, use the child's copying explanation from the overview. For $(1,1)$: "Take from bucket 2 and put one in bucket 1. Now there are two in one bucket, like Problem 1's 2, and the other player goes first. *If needed:* "Does that still work if your partner chooses not to refill?"

Problem 3 (student p. 2): the winning first move

The unique winning results, in printed order across each row, are:

Start	$(1,1)$	$(2,1)$	$(1,2)$	$(3,1)$	$(2,3)$	$(3,3)$
After	$(2,0)$	$(2,0)$	$(0,2)$	$(4,0)$	$(2,2)$	$(4,2)$

Take from bucket 2 except at $(1,2)$, where taking from bucket 1 works. Refill bucket 1 at $(1,1)$, $(3,1)$ and $(3,3)$; do *not* refill at $(2,1)$ or $(2,3)$. Every result is paired, so copying wins thereafter. Any other move leaves an odd bucket and gives the opponent a move to pairs. *If needed:* "What start would you like your partner to face? Is adding a counter always helpful?"

Problem 4 (student p. 3): play the adult

$(4,2)$ favors **2nd**; $(3,4)$ favors **1st**, whose only winning first move is $(3,4) \to (2,4)$, taking from bucket 1. Start with $(4,2)$, child first and then adult first; save $(3,4)$ for a child who explains copying. While the adult plays one challenger, that child's partner checks moves; the other pair works on Problem 5 or replays selected Problem 2 starts. Swap after two games, rather than playing every child through every combination.

Adult play card. When the child moves from paired buckets, quietly take from that same bucket, making the same refill choice. This works at *every* later paired position, not only at the initial one. If it is your turn with an unpaired bucket, take from the highest-numbered unpaired bucket and refill exactly the smaller unpaired buckets; then copy. If all buckets are paired on your turn, no guaranteed win exists: make any legal move and let the child demonstrate the reply. Do not manufacture a win or quietly change a move. Invite children to show what the adult is doing before naming copying. *If needed:* "What changed between your turn and mine? Can you be the adult this time?"

Problem 5 (student p. 3): every move and a reply

From $(2,2)$ the partner has exactly three moves. Each has one winning reply:

After partner's move	After your reply
$(1,2)$	$(0,2)$
$(2,1)$	$(2,0)$
$(3,1)$	$(4,0)$

Completeness: taking from bucket 1 has no refill choice; taking from bucket 2 has two choices, no refill or one in bucket 1. The reply copies that choice and restores pairs. Fill the three rows in any order; each row restarts from $(2,2)$, not the previous row's ending. *If needed:* "Which bucket could your partner take from? Could the same take have a different ending?"

Complete collections and longer games

Problem 6 (student p. 4): every second-player start

Four counters: $(0,4),(2,2),(4,0)$, **three ways**. Six counters: $(0,6),(2,4),(4,2),(6,0)$, **four ways**. Empty buckets are allowed; swapped counts differ because the buckets are numbered. The five slots per row are spare recording space, not a promised answer count. To explain completeness, distribute whole pairs between the two buckets: with two pairs there are three choices for bucket 1; with three pairs there are four. Any split breaking a pair leaves an odd bucket. *If needed:* "Could bucket 1 be empty? Is there another number of whole pairs it could hold?"

Problem 7 (student p. 4): five counters

Impossible. Five counters cannot all be paired inside their buckets: pairs use two at a time, leaving one counter. Therefore every split lets the first player reach pairs. A child may show all six splits, from $(0,5)$ through $(5,0)$, or give the leftover argument. *If needed:* "Where can the last unpaired counter go without making an unpaired bucket?"

Launching the later visit

Switch the goal explicitly. In Problems 8, 9 and 11 the partners cooperate, rather than trying to win. A longest game is not the same question as the largest number in the buckets. At the later visit for Problems 8–12, uncover bucket 3 and set up student p. 1's non-task $(1,0,2)$ on the large board. Children physically make each ending: take from bucket 3, then refill neither bucket, bucket 1 only, bucket 2 only, or both, using Supply. Reset between endings. Each is one whole move under the one-each cap. Then name the cooperative goal for Problems 8, 9 and 11.

Problem 8 (student p. 5): shortest and longest

Read the left column down, then the right column down:

Start	$(0,1)$	$(0,2)$	$(0,3)$	$(1,1)$	$(2,2)$	$(3,3)$
Shortest	1	2	3	2	4	6
Longest	2	4	6	3	6	9

Never refilling achieves the shortest: every original counter must be removed. For the longest, each counter originally in bucket 2 can produce one extra counter in bucket 1; that counter gets one later removal. Thus (a,b) has at most $a+2b$ moves, attained by refilling after every bucket-2 take. A child can match each such take to its one later extra take. *If needed:* "Which takes can give us another turn later? What must happen to every counter already here?"

Problem 9 (student p. 5): most counters held at once

From $(0,0,1),(0,0,2),(0,0,3)$, the answers are **2, 4, 6**. Count only the buckets; pause at the biggest board. Take from bucket 3 and refill both earlier buckets each time to reach $(1,1,0),(2,2,0),(3,3,0)$. Each bucket-3 take can increase the board count by only one; a bucket-2 take cannot increase it, and a bucket-1 take decreases it. There are only one, two or three bucket-3 takes, so these maxima cannot be beaten. *If needed:* "Which take can leave more counters in the buckets than before? How often can we do that?"

Three buckets: strategy and an ending

Problem 10 (student p. 6): winning with three buckets

Top to bottom, the unique winning results are:

Start	$(2,0,1)$	$(1,0,1)$	$(0,1,1)$	$(1,1,1)$
After	$(2,0,0)$	$(2,0,0)$	$(0,2,0)$	$(2,2,0)$

All four take from bucket 3; refill neither bucket, only bucket 1, only bucket 2, then both. These leave pairs, so the same copy works. *If needed:* "Which buckets have a leftover? Can one take and its allowed refills leave none?" Try this when children own the two-bucket strategy.

Problem 11 (student p. 7): longest three-bucket games

Left to right, $(0,1,1), (1,1,1), (0,0,3)$ last at most **6, 7, 12** moves. Refill every allowed bucket to attain each. One bucket-3 counter earns at most four removals: itself, its new bucket-1 counter, its new bucket-2 counter, and that counter's later refill into bucket 1. A bucket-2 counter earns at most two; a bucket-1 counter earns one. *If needed:* "After bucket 3 is empty, are there still ways to earn extra turns?"

Problem 12 (student p. 7): could play last forever?

No. A child can say, "Bucket 3 never gets new counters, so after two takes it is empty for good. Then bucket 2 never gets new ones, so it empties too. Then bucket 1 only goes down. They run out one after another." For adults: between bucket-3 takes only finitely many bucket-2 takes are possible, and between those only finitely many bucket-1 takes. After the last bucket-3 take, the ending two-bucket game remains. A long trial suggests this but does not prove it. Adult check: at most $2 + 2(2) + 4(2) = 14$ moves; counting them is not required. *If needed:* "Which bucket can never get another counter? What game is left after its last take?"

Where it goes (later integration only). After approval, replace the p. 2 paragraph beginning "K–1, adult-led, after the launch" in W81-FAC-v2 with a one-line pointer to this section, placed separately after the existing guide. Change p. 1's "K–1 has an adult-led route on page 2" to name the W81-K-v1 pages and this companion section. Game provenance: Joel David Hamkins, "Buckets of fish!" (2017), as credited in the week's source notes; this K–1 route is an untested teaching adaptation.